Kinematic Differences Between the Dominant and Nondominant Legs During a Single-Leg Drop Vertical Jump in Female Soccer Players

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Background: In soccer, the roles of the dominant (kicking) and nondominant (supporting) legs are different. The kinematic differences between the actions of the dominant and nondominant legs in female soccer players are not clear.

Purpose: To clarify the kinematic differences between dominant and nondominant legs during a single-leg drop vertical jump (DVJ) in female soccer players.

Study Design: Controlled laboratory study.

Methods: A total of 64 female high school and college soccer players were included in this study. Participants performed a single-leg DVJ test utilizing video motion capture with artificial intelligence during the preseason period. This study assessed the knee flexion angles, knee valgus angles, hip flexion angles, and lower leg anterior inclination angle at 3 time points (initial contact, maximum flexion of the knee, and toe-off) and compared them between the dominant and nondominant legs. These angles were calculated from motion capture data and analyzed in 3 dimensions. A paired *t* test was used to analyze the differences between legs, and the significance level was set at $P < .05$.

Results: The knee valgus angle at initial contact was greater in the nondominant leg (mean ± SD, 0.8° ± 5.2°) than the dominant leg (−0.9° ± 4.9°) ($P < .01$). There were no differences between legs for any other angles at any of the time points.

Conclusion: The kinematics of the dominant and nondominant legs of female soccer players in a single-leg DVJ differ in knee valgus angle.

Clinical Relevance: Leg dominance is associated with the risk of sports injuries. Kinematic differences between the dominant and nondominant legs may be a noteworthy factor in elucidating the mechanisms and risk of sports injury associated with leg dominance.

Keywords: drop vertical jump; leg dominance; female; soccer

Soccer is among the most popular sports in the world. In recent years, there has been growing interest in female soccer, and the number of female players is increasing. The playing style of soccer is also changing rapidly, becoming faster and more dynamic.[8] Soccer has a higher incidence of injuries than many other sports.[6,7,13,14,31,37] Approximately 60% to 85% of soccer injuries are to the lower limbs, with the most susceptible joint being the knee, followed by the ankle.[20,36] Anterior cruciate ligament (ACL) injuries and ankle sprains are common injuries for soccer players, and most occur on foot landing and in noncontact cutting.[38] Therefore, it is necessary to manage noncontact sports injuries. Various factors contribute to the risk of noncontact lower limb injuries,[1,2,12,41] and leg dominance is one such factor. Leg dominance can cause asymmetry and unilateral lower limb injuries.[4,11]

Soccer players utilize a range of movements, such as kicking the ball, intermittent dashing, dribbling, turning steps, cutting, and ball manipulation. The kicking leg and supporting leg serve different functions. For instance, the kicking (dominant) leg is swung fast for a powerful kick, and joint and segment velocities must be controlled for an accurate kick.[24] The role of the supporting (nondominant) leg is to attenuate the shock of landing and

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contribute to the acceleration of the swing of the kicking leg.[21] Clarification of the kinematic differences between the dominant and nondominant legs in soccer may usefully contribute to the prevention of injuries, especially for female soccer players. Female soccer players have a higher incidence of noncontact injuries than males,[2] and the movements of the nondominant leg can contribute to the risk of noncontact ACL injury.[3]

A drop vertical jump (DVJ) is a simple and useful means of risk assessment for noncontact ACL injuries,[16] and there have been many studies of the landing kinematics involved in a DVJ. Yet, there has been much controversy over the use of DVJ as an ACL injury predictor.[26] We have been analyzing the dynamics of DVJ as a medical assessment tool in a sports injury prevention project. In this capacity, we analyzed differences in DVJ kinematics between the dominant and nondominant legs of female soccer players. However, there has been little research into the kinematic differences between the dominant and nondominant legs in a single-leg DVJ among female soccer players.

Therefore, the purpose of this study was to identify kinematic differences between the dominant and nondominant legs of female soccer players in a single-leg DVJ. It was hypothesized that the kinematics of a single-leg DVJ leg would differ between the dominant and nondominant legs and that the difference would be greater at initial contact (IC).

METHODS

Study Design and Participants

A total of 64 high school and college female soccer players with a mean age of 16.6 years (range, 15-22 years) were included in this study in 2018 (Table 1). For 57 of these players, the right leg was dominant, and the left leg was dominant in the remaining 7. The kicking leg was defined as the dominant leg, and the supporting leg was defined as the nondominant leg. The inclusion criteria were active soccer players with no injuries at the time of the study and no reported musculoskeletal injuries of the lower limb in the preceding 3 months. This study was a part of a sports injury prevention project: the Prospective Study of Predictors of Sports Injuries: UTokyo Sports Science Initiative. The project includes athletes from other sports and male and female soccer players.[22,23,39] Among them are female soccer players who played competitively on high school and college teams. The sports injury prevention project was approved by the institutional review board of our institution (11907-2). Participants granted their informed consent before participating in the study.

TABLE 1
Participant Characteristics[a]

No. of athletes	64
Dominant leg, right:left	57:7
Age, y	16.6 ± 1.8 (15-22)
Body height, cm	159.0 ± 5.6
Body weight, kg	53.2 ± 5.2
Body mass index	21.0 ± 1.3
Athletic career, y	7.7 ± 3.2 (1-13)

[a]Data are presented as number or mean ± SD (range).

Drop Vertical Jump

The participants were instructed to perform the DVJ as described in previous studies,[16,23] but these instructions were modified so that participants could perform the DVJ with a single leg. First, the participants were to perform a single-leg DVJ on the right foot; second, they performed it on the left foot. The single-leg DVJ was verbally explained to the participants and then demonstrated by instructors. The participants were instructed to stand on 1 leg, barefoot, on the top of a 30-cm box, with their hands free. They were then told to "jump off the box and immediately jump vertically as high as you can." Participants stood on the box with the testing leg, jumped, and landed with the same leg. They immediately performed the second jump as high as possible using the same leg. At each testing session, participants completed 3 to 5 practice trials until they became comfortable with the technique. After completing the practice trials, they took sufficient break time to prevent fatigue. The main trial was then performed only once by each participant.

Dynamic Analysis

This study utilized video motion capture with artificial intelligence (VMocap).[23] The operating procedures and accuracy of VMocap were previously validated.[34] VMocap provides iterative imaging of a human skeletal model of the participant being filmed with measurements of the joint angles calculated from multicamera images. The skeletal model means a virtual open-tree structure kinematic chain. From the RGB images (red, green, and blue) captured by multiple cameras—image data were captured at 30 frames per second on 4 RGB cameras—the person's key joint points can be detected by the artificial intelligence. This is achieved using the convolutional neural network of the OpenPose free software in the form of 2-dimensional part confidence maps.[4,42] There were 18 key

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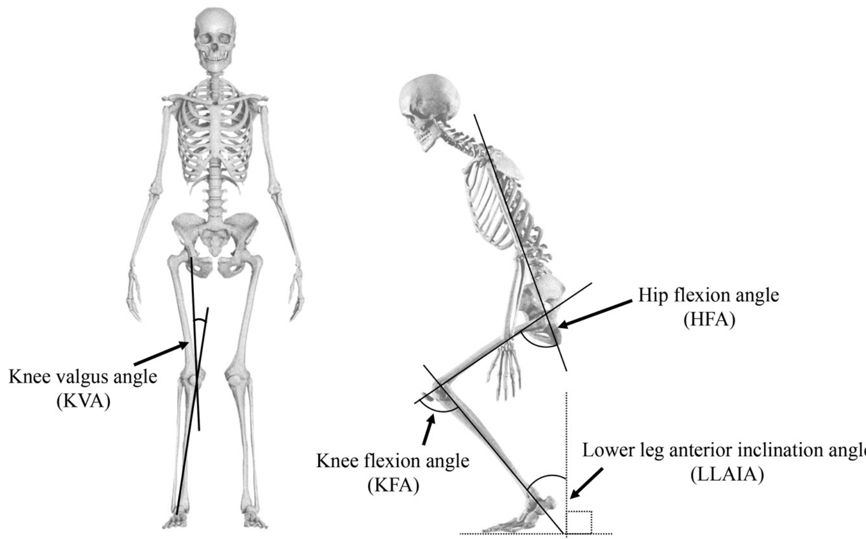

Figure 1. Biomechanical data analysis in the drop vertical jump. Knee valgus angle = angle between the femoral axis and tibial axis in the coronal plane. Hip flexion angle = angle between the trunk and femoral axis in the sagittal plane. Knee flexion angle = angle between the femoral axis and tibial axis in the sagittal plane. Lower leg anterior inclination angle = angle between the tibial axis and a line vertical to the ground.

points that consisted of joint positions at the shoulders, elbows, wrists, hips, knees, ankles, and neck, as well as feature points such as the eyes, ears, and nose. The virtual markers of probable joint positions are generated from these part confidence maps of the key points. Next, inverse kinematic computation minimizes the error between the virtual marker positions and the skeletal model's joint position. This allowed us to analyze each athlete's movements and identify the joint angles throughout the DVJ motions[23] (Figure 1).

Biomechanical Data Analysis

The frontal and sagittal planes were defined by 3 points: the midpoint of the bilateral shoulders and the bilateral hip center points. On the frontal and sagittal planes, the joint angles and inclinations were calculated for the ankle, knee, hip, and chest points of each participant using the motion capture data. The knee valgus angle (KVA) was calculated on the frontal plane, and the knee flexion angle (KFA), hip flexion angle (HFA), and lower leg anterior inclination angle (LLAIA) were calculated on the sagittal plane (Figure 2). These joint and inclination angles were evaluated at the moment of IC with the ground, the point of maximum flexion of the knee, and the toe-off point at which the foot left the ground. The IC and toe-off events were the moments at which the acceleration of the ankle reached zero or left zero.[23]

Statistical Analysis

All statistical analyses were performed using SPSS Version 25 (IBM Corp). A paired t test was used to compare the dominant and nondominant leg angles for the KVA, KFA, HFA, and LLAIA at times IC, maximum flexion of the knee, and toe-off. The significance level was set at $P < .05$. A post hoc power analysis in the KVA at IC was confirmed as 0.74, based on a 2-sided significance level set at $\alpha < .05$, an observed effect size 0.32, and a total sample size of 64.

RESULTS

At IC, the KVA angle was significantly greater in the nondominant legs than the dominant legs of our participants (Table 2). No significant differences between legs were found for any other joint angles at any of the evaluation points.

DISCUSSION

In this study, the differences in dynamics between the dominant and nondominant legs in a single-leg DVJ of female soccer players were clarified using 3-dimensional dynamic analysis. The nondominant legs of our participants showed a significantly greater KVA on landing than the dominant legs. There were no differences between the dominant and nondominant legs for any other joint angles (HFA, KFA, and LLAIA) at any of the evaluation points. To our knowledge, this may be the first study to examine the relationship between leg dominance and KVA on landing in female soccer players in single-leg DVJ.

Regarding the KVA at IC, dynamic valgus is characterized by medial movement of the knee and results from a combination of hip adduction, internal rotation, and external tibial rotation.[18] Muscular control of the lower extremity in the coronal plane alters the KVA. This likely reflects changes in the contraction patterns of the

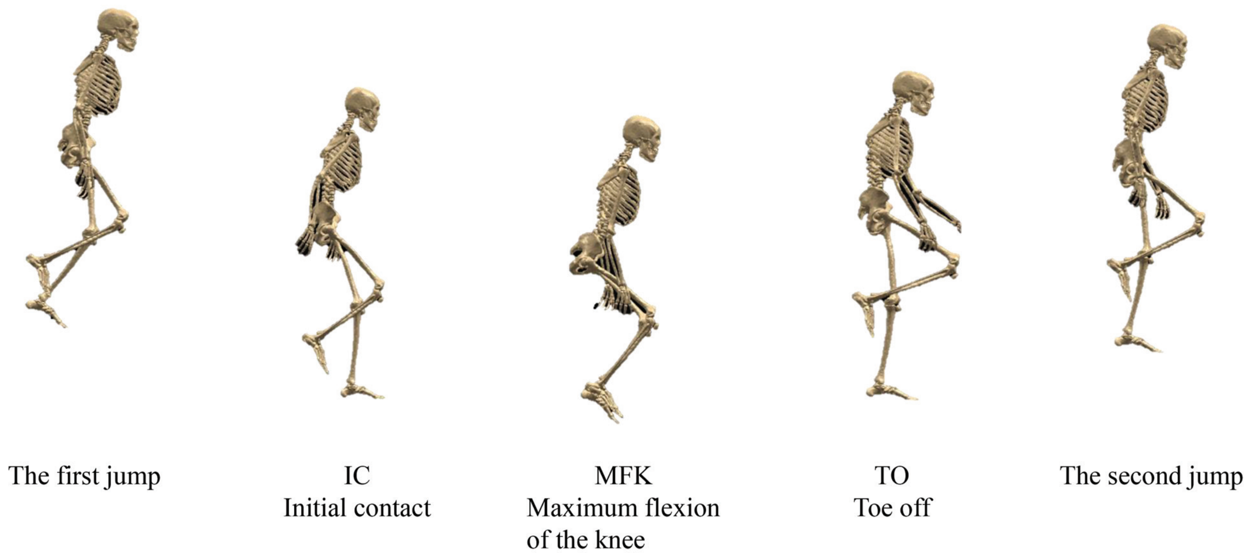

Figure 2. The timing of angle evaluations with video motion capture using artificial intelligence in the drop vertical jump.

TABLE 2
Differences in Range of Motion Between the Dominant and Nondominant Legs at the Point of Initial Contact, Maximum Flexion of the Knee, and Toe-off[a]

Evaluation Point	KVA	KFA	HFA	LLAIA
Initial contact				
Dominant leg, deg	−0.9 ± 4.9	16.3 ± 4.9	21.7 ± 11.2	1.6 ± 3.9
Nondominant leg, deg	0.8 ± 5.2	17.1 ± 5.5	22.9 ± 9.2	1.5 ± 3.4
P value	<.01[b]	.14	.08	.80
Maximum flexion of the knee				
Dominant leg, deg	11.6 ± 11.9	67.4 ± 12.1	52.3 ± 14.9	30.7 ± 5.2
Nondominant leg, deg	10.3 ± 11.4	68.0 ± 11.7	52.9 ± 13.8	31.6 ± 5.1
P value	.40	.45	.54	.09
Toe-off				
Dominant leg, deg	2.1 ± 6.6	20.5 ± 14.7	18.2 ± 11.0	12.8 ± 8.2
Nondominant leg, deg	3.3 ± 6.6	21.3 ± 14.2	19.7 ± 16.0	13.1 ± 7.6
P value	.09	.62	.36	.78

[a]Values are presented as mean ± SD. HFA, hip flexion angle; KFA, knee flexion angle; KVA, knee valgus angle; LLAIA, lower leg anterior inclination angle.
[b]$P < .05$.

adductors and abductors of the knee, primarily the knee flexors, hamstrings, and gastrocnemius muscle, the tendons of which cross the medial and lateral sides of the knee joint.[9] A large valgus angle of the knee upon landing may denote poor control of the coronal knee muscle. However, this study did not evaluate the relationships of muscle strength, muscle control, and balance with the landing kinematics of the dominant and nondominant legs. Therefore, future research should be conducted that includes calculations of muscle strength.

A previous study compared the DVJ of the dominant and nondominant legs of female athletes,[33] although it was not limited to soccer players. The study found that the KVA at IC was greater in the nondominant leg of female recreational athletes but greater in the dominant leg of collegiate athletes.[33] In the current study, the KVA at IC was significantly greater for the nondominant than dominant leg. There are 3 possible reasons for the difference between our findings and those of the previous study: differences in the type of DVJ, the targeted sports, and the definition of the dominant leg. First, the previous study used a DVJ with both legs while the current study used a single-leg DVJ. The single-leg DVJ requires more dynamic balance than the DVJ with both legs. Second, the participants in the previous research were players of basketball, volleyball, or soccer, while those in the present study were all soccer players. Third, although the dominant leg was defined as the kicking leg in both studies,[33] the kicking leg of a basketball or volleyball player may have different functional characteristics from those of a soccer player. The definition of the dominant leg in nonkicking sports should be reconsidered. There are various ways to define the dominant leg. For example, the dominant leg may be the stronger leg,[19] the foot used to initiate stair climbing,[5] the kicking leg,[30,32,40] or the leg used to regain balance after an unexpected perturbation.[10,17]

The KVA on landing is an important index for the risk of a noncontact ACL injury, and the KVA at IC is significantly greater in female athletes who sustain noncontact ACL injuries than those who do not.[34] Additionally, there is a higher risk in female soccer players for noncontact ACL injuries to the nondominant leg.[3,12] It has been speculated that the reason for this may be that the nondominant leg is more likely to be exposed to 1-leg loading patterns associated with ACL injuries, such as shooting, passing, and cutting movements.[12] Landing with a large KVA and using the nondominant leg are known risks for noncontact ACL injuries, but the relationship between KVA on landing and leg dominance is not clear. KVA and valgus moment are predictors of ACL injury, and valgus loading can increase the force on the ACL.[29] Physiological valgus torque on the knee can increase anterior tibial translation and loads on the ACL several-fold.[11] In addition, because ACL injury occurs 0.04 seconds after IC,[25] the lower limb alignment at IC may be important for the risk of noncontact ACL injuries. In this study, although the difference of KVA between the dominant and nondominant legs was small at IC, the KVA of the nondominant leg showed valgus alignment; and, conversely, the KVA of the dominant leg showed varus alignment. As a result, knee abduction moment and dynamic knee valgus loading[15] could increase in the nondominant leg. Therefore, a possible reason why ACL injuries are more common in the nondominant leg may be that the KVA of the nondominant leg is in valgus during IC, contrary to the dominant leg. There was no significant kinematic difference between the dominant and nondominant legs in the HFA, KFA, and LLAIA in the single-leg DVJ in this study. Sagittal plane factors contribute to the ACL injury mechanism,[28] and the single-leg landing technique and posture are related to the risk of ACL injury.[27] Stiff landing—which occurs when there is a high vertical ground-reaction force with lower angles of the ankle, knee, and hip joints—increases the risk of ACL injury.[27]

The HFA and KFA are important indexes for the risk of ACL injury. As the results of this study showed no biomechanical difference in the HFA and KFA between the dominant and nondominant legs on landing, there may be no difference in the single-leg landing technique between the dominant and nondominant legs. The results of this study suggest that it might be better to conduct separate dynamic motion analyses of the dominant and nondominant legs to assess the risk of injuries. The greater KVA of the nondominant leg on landing may be among the reasons for the higher risk of noncontact ACL injuries of the nondominant leg in female soccer players.

There were several limitations to this study. First, only 1 trial was performed per participant; therefore, trial variations may have affected the results. However, we excluded an obvious deviate trial. Nevertheless, >1 trial per participant would have allowed our results to incorporate intrapersonal differences in the features of the single-leg DVJ. Second, the participants were relatively young, as they were all high school and college soccer players. It has been suggested that there is a decrease in neuromuscular control of the knee in female athletes with maturation[15] and that injury risk varies with age and length of athletic career. The present results may not apply to older athletes. Yet, most of our sample had relatively long athletic careers and were members of a top-level team that had recently won a national championship. Third, there is a bias in the number of people with a right or left dominant leg, making the number in the sample with a left-dominant leg rather small. This limitation could be solved by a larger trial. Fourth, the order for each leg at the DVJ was not randomized. The dynamic analysis was performed automatically in the system and so, as such, the order was fixed. Finally, the average estimation error of VMocap is 23.4 mm, which is larger than dynamic motion analyses using markers. However, VMocap enables more imaging and analysis in a shorter time and is easier than dynamic motion analysis with markers.[35] The system may prove beneficial for research into sports injuries and their prevention to be able to obtain and analyze kinematic data from larger numbers of people in a shorter time.

CONCLUSION

A difference was found in female soccer players between the kinematics of the dominant and nondominant legs during a single-leg DVJ. Kinematic differences between the dominant and nondominant legs may be a noteworthy factor in elucidating the mechanisms associated with leg dominance and risk of sports injury.

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